

Fibonacci Numbers, Longest Increasing Subsequence, Knapsack Problem

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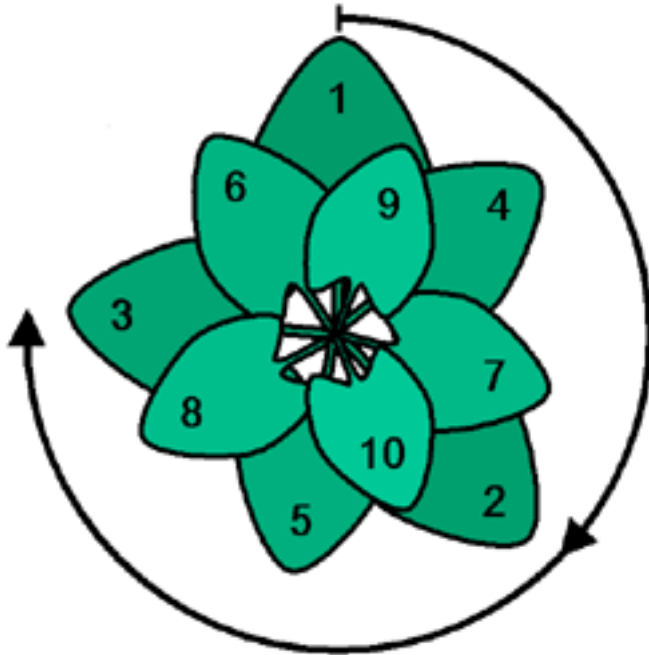
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Fibonacci Numbers

This is a sequence where each number is the sum of the two preceding ones, starting with 0 and 1 (0, 1, 1, 2, 3, 5, 8...). The definition is $F(n) = F(n-1) + F(n-2)$, with base cases $F(0) = 0$ and $F(1) = 1$. You can generate this sequence using a **recursive approach** (a function calling itself) or an **iterative approach** (using loops). Recursion is simple conceptually but highly inefficient for larger inputs with a time complexity of $O(2^n)$. The iterative approach is much more efficient with a time complexity of $O(n)$. Real-life uses include rabbit population modeling, financial algorithms, and approximating the golden ratio.

Plant Branch Growth

A rare plant grows following a Fibonacci-like pattern. In the first cycle, it has 1 branch. In the second cycle, it grows 1 more. From the third cycle onward, the number of new branches equals the total number of branches in the previous two cycles.

Task:

How many branches will the plant have after 12 growth cycles?

Hint:

Model this as a Fibonacci sequence:

- Cycle 1 → 1 branch
- Cycle 2 → 1 branch
- Cycle 3 → 2 branches
- Cycle 4 → 3 branches

Find the total number of branches at cycle 12.

Answer

```
def fibonacci_branches(cycles):
    a, b = 1, 1
    for _ in range(3, cycles + 1):
        a, b = b, a + b
    return b

# Find the number of branches after 12 growth cycles
cycles = 12
branches = fibonacci_branches(cycles)
print(f"After {cycles} growth cycles, the plant will have {branches} branches.")
```

After 12 growth cycles, the plant will have 144 branches.

Longest Increasing Subsequence (LIS)

Given an array of numbers, the task is to find the length of the longest subsequence where the elements are in strictly increasing order. A naive recursive approach exists but has exponential time complexity. More efficient solutions use **Dynamic Programming** (bottom-up tabulation) with $O(n^2)$ time and $O(n)$ space, or **Binary Search** with $O(n \log n)$ time and $O(n)$ space. LIS has applications in areas like stock market analysis, version control systems, and DNA sequence alignment.

Student Score Trends

A student's test scores over a semester are recorded as: [72, 74, 69, 78, 80, 81, 75, 85, 88, 70, 92]

Task:

Determine the longest consecutive sequence of scores where each score is higher than the last one (i.e., a strictly increasing subsequence).

What is the length of this increasing trend?

Answer

```
import numpy as np

def longest_increasing_subsequence(scores):
    counts = []
    for i in range(len(scores)):
        if i == 0 or scores[i] < scores[counts[-1][-1]]:
            counts.append([i])
        else:
            counts[-1].append(i)
    return np.array(scores)[max(counts, key=len)]

# Student scores
scores = [72, 74, 69, 78, 80, 81, 75, 85, 88, 70, 92]

# Find length of the longest increasing subsequence
lis = longest_increasing_subsequence(scores)
print(f"The longest increasing trend is {lis} and the length is {len(lis)}")
```

The longest increasing trend is [69 78 80 81] and the length is 4

Knapsack Problem

Given a set of items, each with a weight and profit, and a bag with a maximum weight capacity (W), the goal is to choose items to put in the bag to **maximize the total profit** without exceeding the capacity. A crucial constraint is that you must take an item completely or not at all (this is the 0/1 Knapsack variant). A naive recursive approach involves considering all subsets of items, but this has exponential time complexity $O(2^n)$. This DP approach has a time and space complexity of $O(n * W)$.

Server CPU Allocation

You manage a server with 30 CPU units available. You have a list of 7 tasks, each requiring a certain number of CPU units and offering a reward in user satisfaction score:

Task	CPU Time	Reward
A	5	30
B	10	40
C	3	20

Task	CPU Time	Reward
D	8	50
E	7	45
F	4	25
G	6	35

Task:

Which combination of tasks should you select to maximize the total reward without exceeding the 30 CPU unit limit?

Answer

```
import pulp
import pandas as pd

# Task data in a DataFrame
df = pd.DataFrame({
    'Task': ['A', 'B', 'C', 'D', 'E', 'F', 'G'],
    'CPU': [5, 10, 3, 8, 7, 4, 6],
    'Reward': [30, 40, 20, 50, 45, 25, 35]
})

# Define binary decision variables for each row
df['Var'] = [pulp.LpVariable(f"x{i}", cat="Binary") for i in df.index]

# Define the problem
prob = pulp.LpProblem("Knapsack", pulp.LpMaximize)

# Objective: maximize total reward
prob += pulp.lpSum(df['Var'] * df['Reward'])

# Constraint: total CPU 30
prob += pulp.lpSum(df['Var'] * df['CPU']) <= 30

# Solve
prob.solve(pulp.PULP_CBC_CMD(msg=False))

# Get selected tasks
selected = df[df['Var'].apply(lambda v: v.value()) == 1]
```

```
print(f"Selected tasks: {selected['Task'].tolist()}")  
print(f"CPU used: {sum(selected['CPU'])}")  
print(f"Reward earned: {sum(selected['Reward'])}")
```

Selected tasks: ['A', 'D', 'E', 'F', 'G']
CPU used: 30
Reward earned: 185

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